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Robot-assisted speaking practice (RASP): Learner affection, engagement, outcomes, and their interplay

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ABSTRACT

Robots are increasingly utilised in language education to physically accompany learners, simulate natural communication through verbal and non-verbal cues, and thereby engage students in various academic tasks, demonstrating potential in improving learner affection, engagement, and language proficiency. So far, research on using robots specifically for developing English as a Foreign Language (EFL) speaking skill, along with associated learner affection and engagement, remains limited, despite the significance of developing EFL speaking skills and setting-specific exploration of learner affection and engagement. To address this gap, we employed a NAO robot to assist 49 Chinese undergraduate students in EFL speaking practice over eight sessions, examining their affection, engagement, outcomes, and the interplay among these factors within the robot-assisted speaking practice (RASP). Data collection included interviews and pre- and post-questionnaires and speaking tests. Findings indicated positive learner affection, deep engagement, and satisfactory EFL speaking proficiency outcomes compared to the baselines. While learner affection significantly enhanced engagement with RASP, neither affection nor engagement had a statistically significant impact on EFL speaking outcomes. Based on these results, we identified the advantages of robots over alternative technologies, peers, and teachers as the interlocutor in assisting speaking practice among adult learners, thereby enriching the robot-assisted language learning framework. We recommend implementing RASP to foster undergraduate EFL speaking development, particularly with a high-quality robot voice and speech recognition system.

1. Introduction

Rapid advancements in robotics have stimulated growing scholarly interest in robot-assisted learning across domains, including language education (Atman Uslu et al., 2023). As educational tools, robots are tangible machines capable of physically interacting with learners (Derakhshan et al., 2024), adopting diverse social roles (van den Berghe et al., 2019), and simulating natural face-to-face communication (Deng et al., 2024), thus supporting a wide range of language pedagogical tasks, such as delivering instructional

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researchers believe that these combined features make NAO a powerful tool for supporting a variety of learning tasks, including EFL speaking practice (Shin & Shin, 2015).

Research has indicated the overall effectiveness of robot assistance in developing language proficiency and explained it through various theories (Derakhshan et al., 2024), such as the theory of immediacy behaviours. Immediacy behaviours refer to verbal and non-verbal communication techniques and actions that demonstrate closeness among people, such as sharing personal anecdotes, offering prompt feedback, praising and encouraging, maintaining eye contact, nodding, and enacting gestures (Finn & Schrodt, 2012). In language learning, this theory argues that teachers' immediacy behaviours can reduce physical and psychological distance between teachers and students, thereby enhancing learners' affection, engagement, and overall language learning efficiency (Hu & Wang, 2023). Robots can perform immediacy behaviours by being physically present with students and employing various verbal and non-verbal communication techniques via synthesised voice, gestures, body movements, and LED eye movements (Derakhshan et al., 2024), thus bridging the psychological and physical distances with learners (Lee & Lee, 2022). Consequently, language learners are more likely to perceive robots as social agents, develop affective attachment to them, and thus have better affection and deeper engagement (van den Berghe et al., 2019). For example, Wang et al. (2013) had 31 elementary learners practise EFL speaking with robots, finding that students, especially the low-proficient ones, felt a connection with the robot and perceived it as a peer. Feeling comfortable interacting with the robot, students demonstrated deep engagement in the practice and achieved satisfactory outcomes (Wang et al., 2013). Hence, robots may improve affection, engagement, and outcomes in language learning by performing immediacy behaviours.

The embodied learning theory has also been used to analyse robot-assisted language learning (Khazaie & Derakhshan, 2024; Pruss, 2022), which emphasises the deep connection between cognitive processes and bodily experiences in learning (Anttila & Svendler Nielsen, 2019; Anttila, 2018). In language learning, the theory posits that multisensory and physical interactions with language learning materials and tasks, such as having face-to-face Q&A sessions in the target language, could enhance cognitive engagement in language learning and thus increase the efficiency of developing language knowledge and skills (Jusslin et al., 2022). From this perspective, robots are highly promising due to their tangibility and ability to display eye gaze, gestures, and body movements, thereby offering embodied experiences of natural language use (Derakhshan et al., 2024). For example, Khazaie and Derakhshan (2024) investigated robot-assisted EFL learning for medical purposes over 15 weeks, where a robot presented virtual patients' histories and needs audibly and visually in EFL and required the students to understand patients' situations and select appropriate medical suggestions. Formative assessments demonstrated the significant effectiveness of robot assistance in developing EFL listening and reading, regardless of learners' language proficiency levels. The researchers explained that robot-assisted language learning provided embodied experiences of applying language knowledge and skills in medical settings, which enhanced cognitive processes supporting comprehension. Hence, robots may improve learner engagement and outcomes in language learning by bringing about bodily experiences of natural language use.

Robot-assisted language learning can also be explained through the affective filter theory (Derakhshan et al., 2024), which highlights the importance of positive learner affection in processing and developing language knowledge and skills (Sheen, 2008). From this perspective, robot-assisted language learning appears promising. Learners have overall positive affection in robot-assisted language learning, exhibiting reduced anxiety, increased comfort, higher self-efficacy, and stronger motivation (Lin et al., 2022), attributed to robots' multimodal, non-judgemental, supportive communicative manners (Lin et al., 2022; Randall, 2019), as well as learners' intrinsic curiosity about robots (van den Berghe et al., 2019). For example, Alemi et al. (2015) found that 30 students learning EFL vocabulary through robot-assisted language learning reported significantly higher levels of confidence, fun, and satisfaction, compared to 16 students using traditional methods, attributing this to the novelty and enjoyment of robot interactions. Alemi et al. (2017) noted similar positive affection among 19 EFL learners after a robot-assisted language learning programme, highlighting the supportive environment created by robots. Hence, robot-assisted language learning may facilitate language development by improving learner affection (Derakhshan et al., 2024).

Supported by these theoretical frameworks, many researchers have argued for robots' overall advantages over 2D technologies in assisting language learning with empirical evidence. For example, in comparison with chatbot-assisted learning, Derakhshan et al. (2025) compared robot-assisted language learning with chatbot-assisted learning in a 16-session programme focused on developing EFL speaking for medical purposes. Learners interacted with a robot in medical role-plays, listening to reports, making medical suggestions, and giving medical-themed commands in EFL. When students' utterances exhibited sufficient accuracy and fluency, the robot would perform actions following their orders. The control group completed the same tasks through oral conversations with a chatbot. The researchers found that robot-assisted language learning was significantly more effective than chatbot-assisted learning in developing EFL speaking for medical purposes and fostering enjoyment in communication, attributing this to robots' greater ability to elicit learner empathy and enhance engagement. Similarly, based on a 15-week experiment, Khazaie and Derakhshan (2024) found a significant advantage of robot-assisted learning over online learning in developing EFL listening and reading skills. Meta-analyses by Lee and Lee (2022) and Wu and Li (2024) also reported a medium-sized average effect of robot-assisted language learning compared to traditional or other technology-mediated learning conditions.

Despite its overall usefulness, robot-assisted language learning does not always lead to significant language development. Its effectiveness may vary across different language skills. Lee and Lee (2022) found that robot-assisted language learning is most effective in developing vocabulary, owing to robots' advantages in delivering language knowledge. Additionally, the type of robot involved in the learning may significantly affect learning outcomes, as different robotic models, features, and functions influence the educational process in distinct ways. For instance, Randall (2019) suggests that anthropomorphic robots, which excel at emulating human communication, may be particularly effective for language learning. Age is also considered an important factor influencing the effectiveness of robot-assisted language learning. Research shows that young learners, such as kindergartners and elementary students,

benefit more from robot-assisted language learning than secondary and adult students due to greater interest and engagement (Derakhshan et al., 2024; Lee & Lee, 2022). The robot's role in language learning activities also influences effectiveness. Robot-assisted language learning is generally more effective when robots function as peers rather than teachers (Lee & Lee, 2022; van den Berghe et al., 2019). This is because learners tend to feel more confident, experience greater social comfort, and develop a stronger sense of relatedness with peer robots, leading to deeper engagement in robot-assisted language learning (Randall, 2019).

2.5. Existing studies and research gaps

Drawing on extensive prior research, we have identified several key factors central to EFL speaking practice and robot-assisted language learning. In terms of EFL speaking practice, it is a crucial means of developing speaking proficiency (Bailey, 2003) and involves dynamics among the learner (*Subject*), speaking tasks (*Object*), and interlocutors (*Community*) (Engeström, 2001a; Harmer, 2007) from the perspective of activity theory. Traditionally, interlocutors are teachers or peers who manage the conversational flow and offer language input and feedback; however, challenges such as large class sizes and interlocutors' language proficiency often constrain this support (Dao, 2020). Two principal factors influence the academic outcomes of speaking practice. One is learner engagement, encompassing behavioural, cognitive, emotional, and social dimensions (Philp et al., 2010), widely acknowledged as vital for language development (Hiver et al., 2020). In the context of EFL speaking practice, deeper engagement leads to better development in speaking proficiency (Zhang et al., 2020). The other principal factor is learner affection. From the perspective of activity theory, learner affection in EFL speaking practice can be analysed from the aspects of the learner (*Subject*), speaking tasks (*Object*), and interlocutors (*Community*) (Engeström, 2001a; Harmer, 2007), comprising self-efficacy (Peng & Woodrow, 2010), social comfort (Archbell & Coplan, 2022), and task perception (Wesely, 2012). Positive affective states promote engagement (Zeinivand et al., 2015) and improvements in speaking proficiency (Mahripah, 2013; Sheen, 2008), although excessive confidence or social comfort may sometimes hinder progress (Klassen, 2006).

In terms of robot-assisted language learning, it involves learners developing language knowledge and skills through interacting with robots (Lin et al., 2022). In this context, robots physically accompany learners and perform immediacy behaviours, assign learning tasks, recognise learner oral output, provide high-quality language input and feedback, and deliver supportive, non-judgemental communication (Randall, 2019). Empirical research has demonstrated the effectiveness of robots in assisting various language learning activities, including speaking practice (e.g., Lee et al., 2011), and improving learner affection, engagement, and language learning outcomes (Derakhshan et al., 2024; Randall, 2019), theoretically grounded in frameworks, such as immediacy behaviour theory, embodied learning theory, and affective filter theory.

Despite a substantial body of research, several gaps persist. Firstly, although developing speaking skills is a fundamental goal in EFL education (Harmer, 2007; Suban, 2021), and robots hold considerable promise in supporting speaking practice and proficiency development, research specifically addressing robot-assisted development of speaking skills remains limited (Derakhshan et al., 2024; Randall, 2019). Existing studies in this direction are mainly for elementary education and presenting mixed findings. For example, Lee et al. (2011) engaged 24 elementary EFL learners in 17 sessions of robot-assisted language learning, where robots provided pronunciation training, feedback, and role-play interactions; their pre- and post-test results indicated significant improvements in speaking proficiency. Conversely, Hong (2016) found no significant gains in speaking proficiency among 25 elementary learners using robot-assisted learning compared to 27 peers in conventional classrooms. Similarly, In and Han (2015) observed no significant improvement in speaking skills and noted potential negative impacts on learner intonation attributed to the robot's unnatural speech patterns. Given the considerable variation in speaking practice topics and complexity across educational levels, alongside these conflicting results, the generalisability of existing findings is limited. To develop a more comprehensive understanding of the effectiveness of robots in assisting speaking practice and developing speaking proficiency, further empirical research appears meaningful, particularly at educational levels beyond elementary, to inform future implementation strategies.

Secondly, investigation of learner engagement in robot-assisted language learning remains limited. Previous studies have shown that robot assistance can improve language learning outcomes by fostering deep cognitive engagement (e.g., Khazaie & Derakhshan, 2024; Shin & Shin, 2015). However, learner engagement is multidimensional, encompassing cognitive, behavioural, emotional, and social aspects that often interrelate (Philp & Duchesne, 2016). It is possible that robot-assisted language learning fosters learner engagement across multiple dimensions, collectively contributing to its effectiveness. Moreover, research has shown that learner engagement may mediate the impacts of learner affection on language learning outcomes (e.g., Mahripah, 2013; Zeinivand et al., 2015), which may apply to robot-assisted language learning. Supporting our hypothesis, Wang et al. (2013) observed that EFL learners experiencing more social comfort in robot-assisted language learning tended to be more engaged in academic tasks and achieve better learning outcomes. Nonetheless, the interplay among learner affection, engagement, and outcomes remains underexplored in robot-assisted language learning. Given the importance of learner engagement in language learning (Philp & Duchesne, 2016), it appears necessary to investigate learner engagement in robot-assisted language learning from different dimensions and examine its interplay with learner affection and outcomes, so as to provide a deeper understanding of robot-assisted language learning and inform its effective implementation.

2.6. Conceptual framework: RASP

To address research gaps, we propose RASP as a promising method to develop EFL speaking proficiency. Grounded in activity theory, RASP comprises three main elements (Fig. 2): (a) the EFL learner (*Subject*); (b) the speaking practice (*Object*), specifically, Q&A sessions (Lin et al., 2022); and (c) a robot and peer learners as interlocutors (*Community*). The three elements interact. The speaking

practice (*Object*) involves the learner (*Subject*), the robot, and peers (*Community*) in EFL Q&A sessions as interlocutors. The robot (*Community*) initiates the practice (*Object*), introduces the task to the learner (*Subject*) (Randall, 2019), and participates in the Q&A sessions (*Object*) by asking questions, sharing personal anecdotes, and encouraging learner output (Deng et al., 2024). Following discussions with peers (*Community*), the learner (*Subject*) answers questions, which are then supported by linguistic scaffolding, and encouragement from the robot and peers (*Community*) (van den Berghe et al., 2019). Eventually, the speaking practice (*Object*) may lead to the development of EFL speaking proficiency (*Outcome*).

RASP may enhance EFL speaking proficiency comprehensively across multiple dimensions, including coherence, fluency, lexical resource, grammatical range and accuracy, and pronunciation (Chen, 2025), by fostering learner affection and engagement, while controlling for learners' baseline levels (Fig. 3).

Regarding learner affection, robots can create a supportive, non-threatening, and enjoyable environment for speaking practice (Lin et al., 2022; Randall, 2019) by enabling multisensory communications (Alemi et al., 2015, 2017) and creating physical and mental immediacy (Lee & Lee, 2022). Positive learner affection in RASP is reflected on three dimensions: (a) social comfort (Archbell & Coplan, 2022), reflecting learners' ease and confidence in interactions with peers and robots (*Community*); (b) EFL speaking self-efficacy (Peng & Woodrow, 2010), reflecting learner confidence (*Subject*) in practice; and (c) task perception (Aubrey et al., 2022), concerning motivation, interest, and perceived usefulness of the practice (*Object*). According to affective filter theory (Sheen, 2008), learners with better affection in RASP process language knowledge and skills more efficiently, thereby achieving better EFL speaking development.

Regarding learner engagement, it reflects on four dimensions in RASP: (a) behavioural engagement (Gobert et al., 2015), such as listening to and answering the robot's questions; (b) cognitive engagement (Plass et al., 2015), involving the recall of language knowledge and response planning; (c) emotional engagement (Philp & Duchesne, 2016), including sense of belonging and enjoyment during practice; and (d) social engagement (Svalberg, 2009), encompassing interactions with the robot and peers. Robots can provide bodily experiences of face-to-face EFL speaking practice and thus foster deep learner engagement in speaking practice, following the embodied learning theory (Derakhshan et al., 2024). Moreover, robot assistance can further enhance learner engagement by fostering positive learner affection, following the theory of planned behaviour (Ajzen, 2005) and the expectancy-value theory (Wigfield & Eccles, 2000). Deeper engagement in RASP should lead to better outcomes in terms of EFL speaking proficiency, aligning with the engagement theories (Hiver et al., 2020) and the noticing hypothesis (Schmidt, 2012).

By developing, implementing, and investigating RASP among EFL university students, this study aims to elucidate the mechanisms through which robots assist with speaking practices as the interlocutor and facilitate speaking proficiency development via learner affection and engagement at the higher educational level. In doing so, the study may contribute to a more comprehensive understanding of the application and effectiveness of robots in assisting EFL speaking practice and proficiency enhancement (Derakhshan et al., 2024; Randall, 2019). Furthermore, we explored the multidimensional nature of learner engagement, including cognitive, behavioural, emotional, and social dimensions, and its interaction with learner affection and learning outcomes within robot-assisted language learning contexts, an essential area that remains underexplored. Ultimately, this research aims to generate evidence-based insights to inform the design and effective implementation of RASP interventions and advance theoretical understanding of robot-assisted language learning and EFL speaking development.

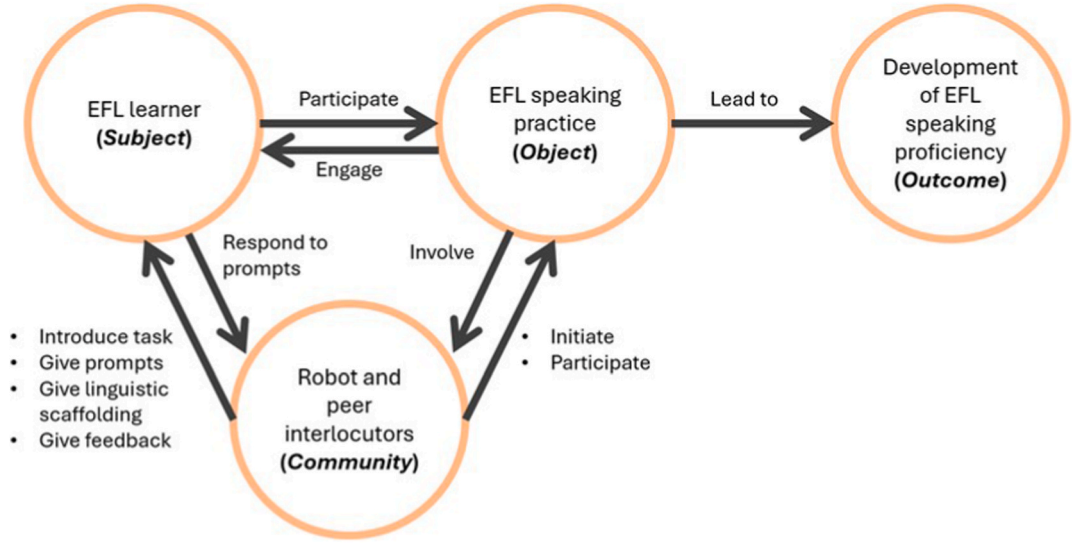

Fig. 2. Elements of RASP in light of the activity theory.

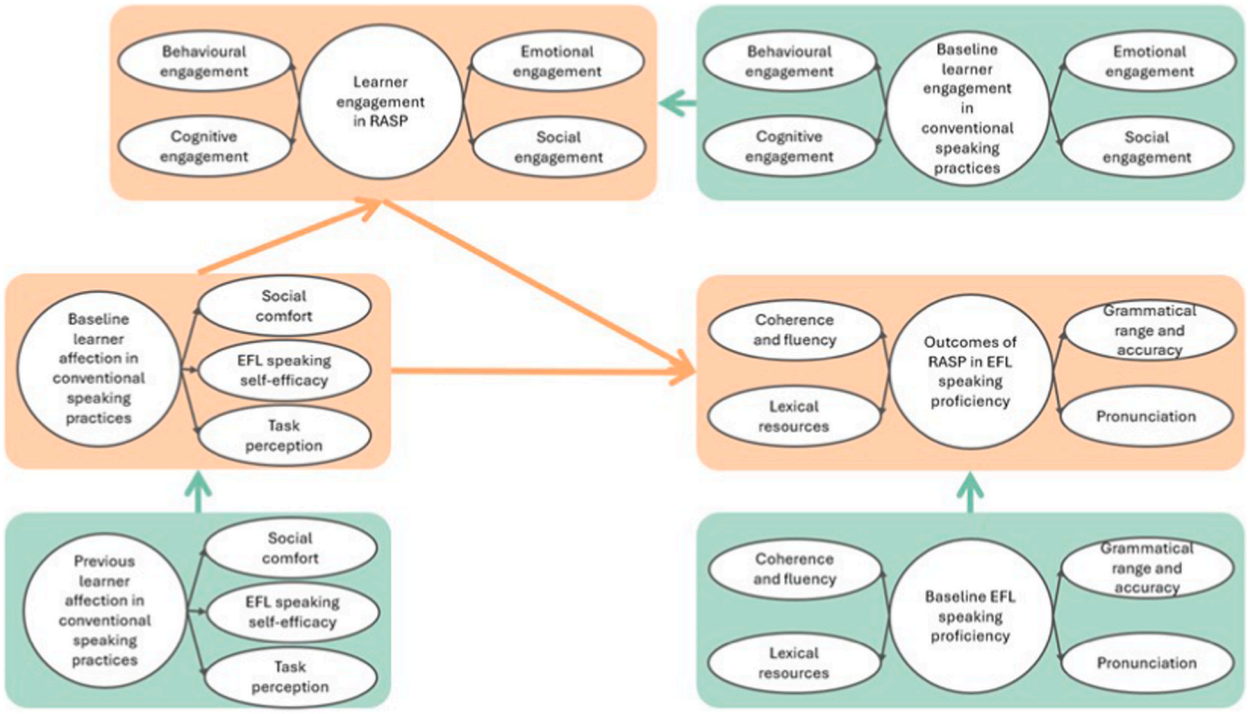

Fig. 3. Underlying mechanism of the effectiveness of RASP.

3. Method

We adopted a concurrent triangulation mixed-methods approach, simultaneously collecting, analysing, and interpreting skill test scores, questionnaire responses, and interview transcripts for cross-validation and corroboration (see Riazi & Farsani, 2023). To control for baseline factors, we administered pre- and post-questionnaires and tests to examine learner affection, engagement, and outcomes of RASP.

3.1. Participants

Through convenience sampling, we recruited a total of 50 undergraduate Chinese students in a university of Hong Kong. All students were required to achieve an IELTS score of 6.5 for graduation and had voluntarily enrolled in an English-medium course designed to prepare them for the IELTS test, indicating strong motivation to enhance their EFL proficiency. Students completed an online pre-treatment survey collecting demographic information, their needs for developing EFL speaking proficiency, and their perceptions and prior experiences with technology-enhanced learning, EFL speaking practices, and robot-assisted learning.

Based on the survey, all students reported strong needs for developing EFL speaking proficiency and some experiences with technology-enhanced learning and conventional EFL speaking practices, including self-regulated, peer-based, or teacher-led methods. None had prior experience with robot-assisted learning. After excluding one student who reported technophobia, the final cohort comprised 49 participants (22 females and 27 males), aged 18 to 23.

Before the intervention, each participant signed a consent form affirming that (a) their personal information would remain confidential, (b) their performance and participation would not affect their academic standing, and (c) they could withdraw from the study at any time without consequence.

3.2. Instruments

3.2.1. Practice materials

We developed 50 sets of practice materials, each consisting of one Q&A session (Lin et al., 2022) and linguistic scaffolding (Deng et al., 2024). Each Q&A session included three authentic IELTS speaking questions sourced from the British Council website (https://takeielts.britishcouncil.org/take-ielts/prepare/free-ielts-english-practice-tests/speaking). Each session focused on a specific theme (e.g., the use of technology) to establish a coherent context for speaking activities, as recommended by Lin et al. (2022). The questions were sequenced from lower to higher levels of complexity, aiming to scaffold learners and facilitate steady progress. For example:

- Question One: "What kind of technology do you often use in your daily life?"

- Question Two: "How has technology changed the way people communicate?"
- Question Three: "Do you think rapid technological development could negatively impact society? Why or why not?"

The linguistic scaffolding was aligned with the Q&A session. Two experts with over five years of IELTS teaching experience reviewed all question sets. For each question, they selected two vocabulary words, phrases, or sentence structures to enhance learners' responses. For instance, for the question regarding negative societal impacts, they recommended the terms "exacerbate" and "detrimental effects." The first author then developed instructional explanations and usage examples for each term, consulting the Longman Dictionary of Contemporary English Online (https://www.ldoceonline.com/). The following is an example instruction of "detrimental effects":

To further enhance your answer, you may include the phrase "detrimental effects," which refers to harmful or damaging consequences. For example, you could say, "The rapid development of technology could have detrimental effects on employment and privacy in the future."

Two experts in IELTS preparation and EFL education subsequently validated the reliability and suitability of the linguistic scaffolding for supporting university students' EFL academic speaking development.

3.2.2. NAO: Robot for assisting speaking practices

We employed a NAO robot to implement RASP due to NAO's evidenced usefulness in supporting EFL learning activities (e.g., Shin & Shin, 2015). Supporting synthesised voice and speech recognition, NAO can serve as the robot interlocutor (*Community*) in RASP that initiates and participates in the practice (*Object*) and interacts with learners (*Subject*) by introducing academic tasks, asking questions, encouraging learner output, and giving feedback and linguistic scaffolding (Deng et al., 2024; Lin et al., 2022; Randall, 2019). These functionalities directly contribute to the development of EFL speaking proficiency (*Outcome*), aligning with our proposed RASP framework.

Furthermore, NAO's capabilities align with the theoretical framework of RASP. NAO's human-like torso, articulated limbs, LED eyes, and multimodal communication systems facilitate multisensory interaction and bodily speaking experiences, thereby enhancing learner engagement and outcomes following embodied learning theory (Khazaie & Derakhshan, 2024; Pruss, 2022). Its capacity to perform complex physical gestures and non-verbal social cues affords both physical and psychological immediacy (Randall, 2019), which, consistent with immediacy behaviour theory (Finn & Schrodt, 2012), fosters improved learner affection, engagement, and academic performance. Additionally, NAO can support multimodal, supportive, and non-judgemental communicative style following pre-programmed models, so as to stimulate learner curiosity and positive affection (Shin & Shin, 2015; Valagkouti et al., 2022), further enhancing learning outcomes as explained by affective filter theory (Derakhshan et al., 2024). Collectively, these features position NAO as a highly effective tool for implementing the RASP framework.

Using Choregraphe, the official programming software for NAO, we designed and executed NAO's interactive behaviours for each RASP session. Each RASP session, lasting approximately 15 min, was based on one of the 50 practice material sets. For each session, NAO greeted the students and introduced the task. Upon receiving affirmative responses such as "Ready" or "OK," NAO initiated the Q&A by sequentially asking three questions and encouraging learners to discuss their answers with peers ("You may discuss the answer with peers."). After recognising learner responses, NAO acknowledged their contributions ("Wow! You made a very good point!"), provided immediate feedback ("Your answer is very good!") and encouragement ("Well done!"), and related personally to their answers ("Like you, I also want to study abroad, perhaps majoring in classical music."). Additionally, NAO offered linguistic scaffolding and prompted students to apply the new vocabulary or structures in a subsequent attempt ("Please try to answer the question again, using the new word!"). Once all three questions had been asked and answered twice, NAO concluded the session. Throughout the interaction, NAO used synthesised speech, gestures, body movements, and LED signals in a vivid and flexible manner.

3.2.3. Pre- and post-tests on EFL speaking proficiency

To evaluate RASP outcomes while controlling for baseline levels, we conducted pre- and post-tests of learners' EFL speaking proficiency. Following the IELTS test format, participants completed individual face-to-face speaking assessments lasting approximately 10 min with the first author as examiner, who possesses extensive experience in assessing IELTS speaking tests. During the assessment, participants responded to three questions without access to reference materials. All questions were sourced from an authentic IELTS question bank (https://ieltsbank.com/product/ielts-questionbank-premium), ensuring high reliability, consistency, and validity. Participants' performances were video-recorded with their consent.

3.2.4. Pre- and post-questionnaires on learner affection

To assess learner affection in RASP while controlling for baseline factors, we administered a pre-questionnaire measuring affection in conventional EFL speaking practices and a post-questionnaire focusing on RASP. The questionnaires used a 10-point Likert scale, chosen for its accuracy, reliability, and suitability for advanced analytical techniques like Partial Least Squares Structural Equation Modelling (PLS-SEM) (Dawes, 2008). The questionnaire comprised 17 items across three dimensions of learner affection identified in our literature review: social comfort, EFL speaking self-efficacy, and task perceptions (Appendix A).

Items 1–3 addressed social comfort and were adapted from Connor et al.'s (2001) abbreviated Social Phobia Inventory scale. For example: "In EFL speaking practice, fear of embarrassment causes me to avoid doing things or interacting with peers."

Items 4–12 assessed EFL speaking self-efficacy and were adapted from Gan et al.'s (2022) scale on self-efficacy in EFL performance among Chinese university students (Cronbach's alpha $\geq$0.86). An example item is: "I can speak English fluently in EFL speaking practices."

Items 13–17 focused on task perceptions and were adapted from Ghani et al.'s (2019) validated scale (Cronbach's alpha = 0.85). An example item is: "Developing English speaking via EFL speaking practices is a good idea."

3.2.5. Pre- and post-questionnaires on learner engagement

To measure learner engagement with RASP while controlling for baseline factors, we administered a pre-questionnaire on learner engagement in conventional EFL speaking practices and a post-questionnaire focusing on RASP. The questionnaire consisted of 12 items rated on a 10-point Likert scale, covering four dimensions of learner engagement: behavioural engagement, cognitive engagement, emotional engagement, and social engagement (Appendix B).

Behavioural engagement (Items 1–3) and cognitive engagement (Items 4–6) items were adapted from Reeve (2013) (Cronbach's alpha >0.84). An example item is: "In EFL speaking practices, I work as hard as I can."

Emotional engagement (Items 7–9) and social engagement (Items 10–12) items were adapted from Deng et al. (2020) (Cronbach's alpha >0.7; factor loadings >0.70; AVE >0.55). An example item is: "I find EFL speaking practices interesting."

3.2.6. Interview questions

We conducted semi-structured interviews based on four groups of questions aligned with the research questions:

RQ1:

- How do you feel about RASP? Why would you have such feelings?

RQ2:

- To what degree do you feel engaged in RASP? Why would you have such feelings?

RQ3:

- Do you believe your EFL speaking skills have improved through RASP? Why would/wouldn't you believe so?

RQ4:

- How do you think your affection towards RASP influences your engagement and EFL speaking development? Please explain the reason(s).
- Does your engagement in RASP influence your EFL speaking development? Why or why not?

These interview questions were refined and validated through a pilot study involving five students and expert recommendations, confirming their credibility.

3.3. Procedures

The experiment was conducted over eight weeks (Fig. 4). In Week One, we collected and analysed students' biographical survey responses to select eligible participants. After obtaining signed consent forms, we conducted a 15-min orientation covering EFL speaking practices, robot-assisted language learning, learner affection, and engagement. Participants then completed pre-

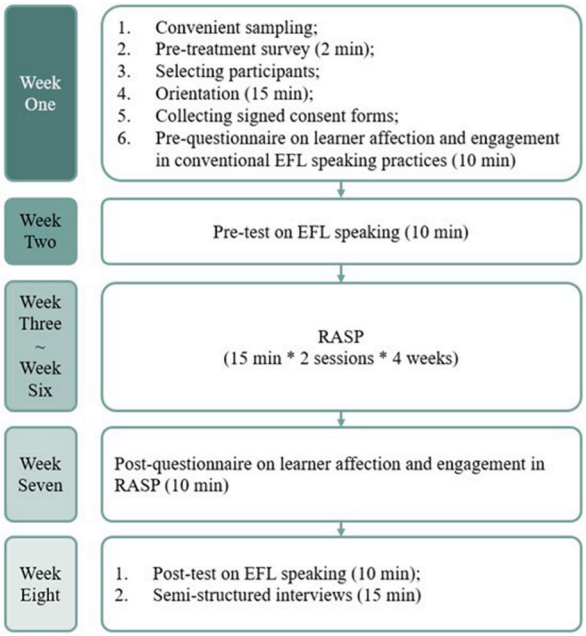

Fig. 4. Experimental workflow.

questionnaires on their affection and engagement in conventional speaking practices (approximately 10 min).

In Week Two, participants completed the 10-min pre-test on EFL speaking.

From Week Three to Week Six, participants attended two 15-min RASP sessions per week (eight sessions in total) following Randall's (2019) recommendations. Organised in groups of two to four, the students practiced speaking by interacting with NAO and group members (Fig. 5). A research assistant provided technical support throughout.

In Week Seven, participants spent approximately 10 min completing the post-questionnaires on affection and engagement in RASP.

In Week Eight, participants completed a 10-min EFL speaking post-test. Two days later, ten participants were randomly selected for 15-min semi-structured interviews conducted in conversational Chinese, their native language. We briefly re-explained the concepts of learner affection and engagement before asking participants to reflect on their RASP experiences and respond to specific questions.

3.4. Data analyses

3.4.1. Qualitative analysis

The authors collaboratively transcribed and analysed the interview data. We repeatedly reviewed the transcripts to familiarise ourselves with the content and developed initial codes aligned with the research questions:

- RQ1: Learner perceptions and reasons for affection in RASP.
- RQ2: Learner perceptions and reasons for engagement in RASP.
- RQ3: Learner perceptions and reasons for EFL speaking proficiency outcomes from RASP.
- RQ4: Learner perceptions and reasons for the interplay between affection, engagement, and EFL speaking proficiency outcomes.

The research team analysed the transcripts using the established coding scheme, noting the frequency of key ideas and identifying representative quotations. We compared coding results and resolved discrepancies through discussion. Finally, we collaboratively translated the coding outcomes and quotations into English.

3.4.2. Qualitative rigour

We applied several techniques to ensure the trustworthiness of the qualitative part of our study, following Shenton (2004).

Before the interviews, we piloted the interview questions with five students and revised them based on expert recommendations to confirm their clarity and alignment with the research aims.

Before data analysis, the research team critically reflected on their backgrounds and potential biases as experts in technology-enhanced language learning and fluent Chinese speakers. This reflection heightened our awareness of subjective interpretations that might overemphasise RASP's positive impacts while underrecognising negative feedback or challenges reported by participants.

During data analysis, the research team jointly coded three transcripts to develop and agree upon a consistent coding scheme.

Fig. 5. Students engaging in RASP.

Following the scheme, we individually analysed the remaining transcripts, noting the frequency of key ideas and identifying representative quotations. Individual coding results were compared, and discrepancies were resolved through discussion, ensuring interpretative consistency. The procedures for data collection, transcription, coding, and analysis were documented for tracking.

After data analyses, interviewees were invited to review and confirmed the results of our analyses, validating the data accuracy.

Finally, we provide thick descriptions of the participants, interview context, and the semi-structured interview protocol, allowing readers to assess the applicability of findings to similar educational settings.

Collectively, these methodological strategies ensured the qualitative components of the study adhered to rigorous standards, strengthening the validity and reliability of insights into learners' experiences with robot-assisted speaking practice.

3.4.3. Quantitative analysis

This study involved two types of quantitative data: test scores and survey responses. All 49 participants completed the pre-test, while one participant missed the post-test. The first author and an experienced IELTS instructor independently scored participants' video-recorded performances based on IELTS criteria: coherence and fluency, lexical resource, grammatical range and accuracy, and pronunciation, each rated from 1 (minimal) to 9 (expert). The comparison of scores revealed high inter-rater reliability (Cohen's Kappa = 0.80), with any differences resolved through discussion.

For the questionnaire surveys, the authors collaboratively cleaned the data by removing incomplete and straight-lining responses, following Meade and Craig (2012). After cleaning, 43 valid pre- and post-questionnaire responses remained for social comfort and EFL speaking self-efficacy. For task perceptions and learner engagement, 41 pre-questionnaire and 43 post-questionnaire responses were valid.

To address RQs 1 to 3, we conducted descriptive analyses focusing on means and standard deviations of questionnaire and test scores. Further investigation involved paired-sample t-tests via SPSS to identify differences between RASP and baselines in EFL speaking proficiency, learner affection, and engagement.

For RQ4, we performed PLS-SEM using the SEMinR package in R (R Core Team, 2019) to investigate relationships within our proposed two-level RASP model (Appendix C, Table 1). Level 1 comprised three primary variables with corresponding baseline controls: (a) "Learner affection in RASP," controlled by "Baseline learner affection in conventional speaking practices"; (b) "Learner engagement in RASP," controlled by "Baseline learner engagement in conventional speaking practices"; and (c) "Learning outcomes of RASP in EFL speaking proficiency," controlled by "Baseline EFL speaking proficiency." "Learner affection in RASP" influences "Learning outcomes of RASP in EFL speaking proficiency" directly and indirectly via "Learner engagement in RASP." At Level 2, all Level-1 variables were reflected by indicators identified in the literature review, measured through questionnaire or test results. Path coefficient significance was assessed using standard bootstrapping with 10,000 subsamples (two-tailed, $\alpha = .05$).

To further clarify RQ4, we conducted multiple linear regression analyses in SPSS to evaluate the predictive power of relationships identified as significant by the PLS-SEM analysis.

3.4.4. Quantitative rigour

Our sample size was sufficient according to rules of thumb: (a) at least 30 observation pairs for paired-sample t-tests; (b) at least ten times the largest number of indicators per variable for PLS-SEM; and (c) at least ten observations per predictor for multiple linear regressions.

The data demonstrated linearity and homoscedasticity, evidenced by the random distribution of points in scatterplots and alignment along the diagonal in P-P plots (Field, 2018). Skewness (−1.94 to 0.83) and kurtosis (−6.81 to 1.79) values confirmed data normality (Byrne, 2010), verifying the suitability of our data for paired-sample t-tests and multiple linear regression analyses (Field, 2018). Additionally, the data were appropriate for PLS-SEM analysis, which does not require adherence to specific distributional assumptions (Hair Jr et al., 2021).

No issues of multicollinearity were identified, as Pearson correlation coefficients between all pairs of indicators were below 0.88, confirming data suitability for multiple linear regression (Field, 2018).

For the PLS-SEM measurement model, we assessed validity and reliability according to Hair Jr et al.'s (2021) guidelines (Appendix C, Tables 2–4). The model exhibited strong internal consistency reliability (rho A > 0.70), adequate discriminant validity (HTMT <0.90), robust convergent validity (AVE >0.50), absence of collinearity issues (VIF <5.00), and acceptable indicator reliability (most loadings exceeding 0.708). Although loadings for "Baseline EFL speaking proficiency" and "Behavioural engagement" and "Cognitive engagement" as indicators of "Baseline learner engagement in conventional speaking practices" were slightly low (between 0.40 and 0.70), these indicators were retained due to the model's overall high reliability and validity.

Data analyses were independently conducted by members of the research team, resulting in excellent inter-rater reliability (Cohen's Kappa = 0.95). Any discrepancies were resolved through collaborative discussions.

4. Results

4.1. Learner affection in RASP

On a 10-point scale (1 = very low/negative; 10 = very high/positive), learners reported relatively high social comfort during RASP ($M = 7.08$, $SD = 2.42$), exceeding the baseline level ($M = 5.74$, $SD = 2.22$); moderately high EFL speaking self-efficacy during RASP ($M = 6.29$, $SD = 1.69$), above the baseline ($M = 5.09$, $SD = 1.63$); and overall positive task perceptions towards RASP ($M = 6.39$, $SD = 1.99$), slightly below baseline ($M = 6.67$, $SD = 1.54$) (Fig. 6).

Paired-sample t-tests (Table 1) indicated that learners' social comfort in RASP was significantly higher than baseline, $t(38) = 3.58$, $p < .001$, whereas EFL speaking self-efficacy and task perceptions did not differ significantly.

Triangulated with quantitative results, our interviews showed that most participants felt positive affection in RASP, highlighting improved social comfort and EFL self-efficacy. They attributed these improvements to NAO's non-judgemental attitudes and consistent encouragement, which alleviated their usual nervousness during speaking activities. One participant stated, "NAO kept encouraging me like a friend … I felt more relaxed when speaking English to NAO." Additionally, interviewees reported that RASP helped them focus on the task, reducing concerns about personal language limitations or peer judgements. One participant remarked, "I usually feel nervous about my speaking accuracy when talking in English with peers … I don't want to sound foolish. But [in RASP], I focused entirely on answering NAO's questions and forgot how my classmates might judge my speaking."

However, some students reported negative affection in RASP, mainly due to NAO's technical limitations. Several interviewees cited instances where NAO misrecognised their speech and led to delays or interruptions, which made them feel "anxious" and "reluctant". NAO's synthesised voice also posed comprehension challenges, with some interviewees describing it as "strange and incomprehensible," which negatively impacted their task perceptions of RASP.

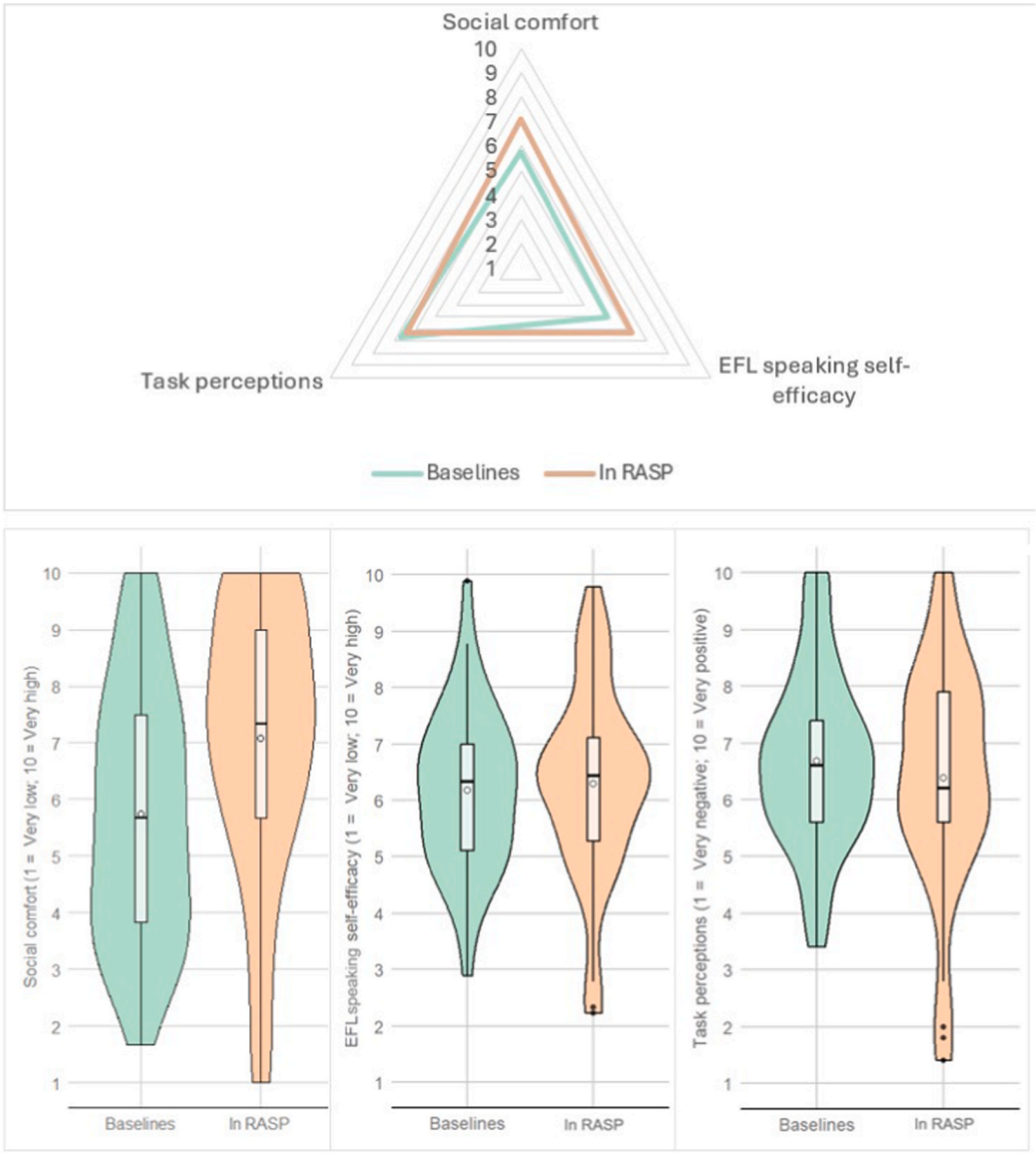

Fig. 6. Learner affection in RASP against the baselines.

Table 1
Paired-sample t-tests on the differences between learner affection in RASP and the baselines.

	M	*SD*	95% *CI*	*t*	*df*	Two-Sided *p*
Social comfort	1.23	2.16	(-1.94-, 0.53)	3.578	38	<0.001**
EFL speaking self-efficacy	1.32	5.33	(-0.40, 3.05)	1.552	38	0.129
Task perceptions	−0.37	2.29	(-1.14, 0.39)	−0.990	36	0.329

**$p < .001$.

4.2. Learner engagement in RASP

On a scale from 1 (very low) to 10 (very high), learners reported high behavioural engagement ($M = 7.20$, $SD = 1.63$) and moderately high cognitive ($M = 6.78$, $SD = 1.54$), emotional ($M = 6.49$, SD = 2.06), and social engagement ($M = 6.10$, $SD = 1.93$) with RASP (Fig. 7). By comparison, baseline scores for conventional speaking practice were lower in behavioural ($M = 6.91$, $SD = 1.37$), cognitive ($M = 6.50$, $SD = 1.32$), and social engagement ($M = 5.71$, $SD = 1.79$), but slightly higher emotional engagement ($M = 6.59$, $SD = 1.47$).

Paired-sample t-tests (Table 2) revealed no significant differences in learner engagement between RASP and the baseline.

Interviewees generally confirmed high engagement in RASP, often citing the perception of "being heard and responded to." They highlighted NAO's attentiveness, timely feedback, and interactive gestures as factors that enhanced motivation and concentration. One participant noted, "Usually, peers pay little attention … However, NAO listened carefully, which really encouraged me to try harder." Another interviewee commented, "NAO is like a friend willing to listen. When NAO was looking at me and nodding, I felt I was heard and understood."

Additionally, learners expressed excitement about using NAO as a novel learning tool, with one stating, "I've never learnt with a robot before, it's very exciting." However, most participants acknowledged that their initial interest waned after two to three sessions, with engagement levels "returning to those typical of conventional speaking activities".

4.3. RASP outcomes in terms of EFL speaking proficiency

On a scale from 1 (minimal) to 9 (expert), learners exhibited moderately high speaking proficiency levels after RASP: coherence and fluency ($M = 6.58$, $SD = 1.05$), lexical resource ($M = 6.31$, $SD = 0.72$), grammatical range and accuracy ($M = 6.62$, $SD = 0.73$), and pronunciation ($M = 6.46$, $SD = 0.71$) (Fig. 8). In comparison, baseline proficiency levels were lower: coherence and fluency ($M = 5.86$, $SD = 1.10$), lexical resource ($M = 6.04$, $SD = 0.58$), grammatical range and accuracy ($M = 5.80$, $SD = 0.74$), and pronunciation ($M = 6.14$, $SD = 0.65$).

Paired-sample t-tests (Table 3) indicated significant improvements in EFL speaking proficiency as a result of RASP, in coherence and fluency ($t(47) = 6.40$, $p < .001$), lexical resource ($t(47) = 2.72$, $p = .009$), grammatical range and accuracy ($t(47) = 9.08$, $p < .001$), and pronunciation ($t(47) = 3.67$, $p < .001$).

Corroborating the test results, all interviewees reported improvements in their speaking proficiency, attributing these gains to the semi-formal, real-time interactions with NAO and the high-quality language input that NAO offered. Participants noted that NAO's questioning, feedback, and language delivery were "very professional" and "close to authentic IELTS speaking examiners". The robot's physical presence enhanced its role as a "real interlocutor," prompting learners to take practice "more seriously" than peer-based or self-regulated speaking activities. One student remarked, "I took the practice with NAO as seriously as practicing with my English teacher."

Moreover, several participants highlighted NAO's effectiveness in conveying language knowledge, noting that its "engaging hand gestures" and "eye contact" made language knowledge "memorable" and "easy to understand." However, the quality of NAO's synthesised voice posed comprehension challenges, with one stating, "Sometimes, NAO's pronunciation can be quite strange and hard to understand. I cannot really learn from it."

4.4. Interactions among learner affection, engagement, and outcome of RASP

Our PLS-SEM analysis, based on Hair Jr et al.'s (2021) criteria, demonstrated moderate-to-large predictive power for the proposed model, explaining significant variance in learner affection ($R^2 = 0.31$), engagement ($R^2 = 0.74$), and EFL speaking proficiency outcomes ($R^2 = 0.58$) of RASP (Fig. 9). Significant paths are: "Learner affection in RASP" to "Learner engagement in RASP" ($\beta = 0.86$, $t = 10.62$, 95% *BCCI* = [0.68, 1.00], $f^2 = 1.76$); "Baseline learner affection in conventional speaking practices" to "Learner affection in RASP" ($\beta = 0.55$, $t = 5.56$, 95% *BCCI* = [0.35, 0.74], $f^2 = 0.44$); and "Baseline EFL speaking proficiency" to "Outcomes of RASP in EFL speaking proficiency" ($\beta = 0.78$, $t = 11.51$, 95% *BCCI* = [0.63, 0.90], $f^2 = 1.22$). These results suggest that learners with positive affection in conventional speaking practices maintained favourable affection in RASP, which in turn significantly enhanced their engagement. Additionally, baseline proficiency levels were strong predictors of improved speaking proficiency after RASP.

To further explore the significant link between learner affection and engagement identified by PLS-SEM, we conducted multiple linear regressions. Results (Appendix D) revealed significant effects of learner affection on all four dimensions of learner engagement in RASP: behavioural ($R^2 = 0.66$, $F(3, 39) = 25.55$, $p < .001$), cognitive ($R^2 = 0.59$, $F(3, 39) = 18.59$, $p < .001$), emotional ($R^2 = 0.81$, $F(3, 39) = 54.72$, $p < .001$), and social engagement ($R^2 = 0.46$, $F(3, 39) = 11.02$, $p < .001$). Power analyses conducted with G*Power

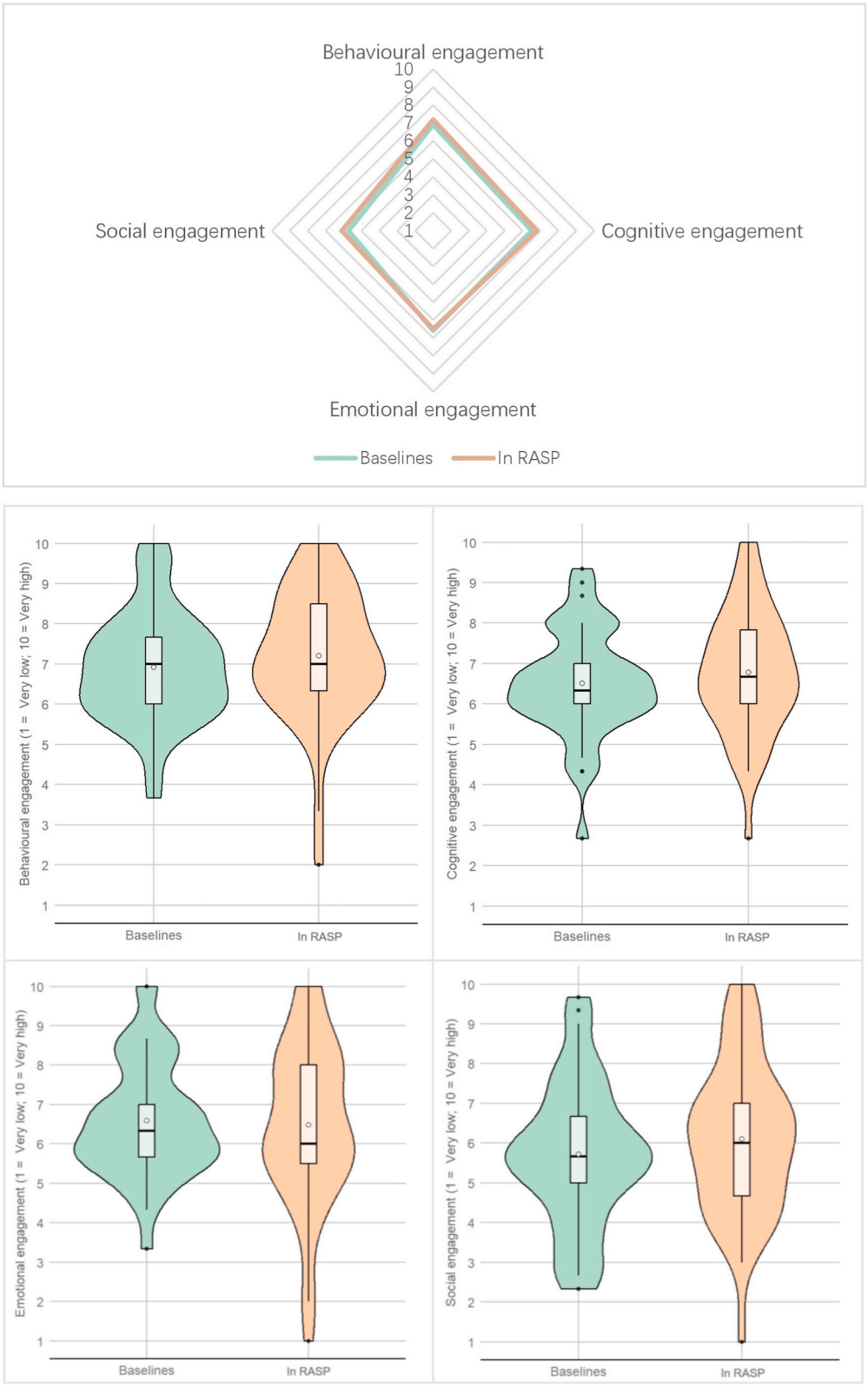

Fig. 7. Learner engagement in RASP against the baseline.

Table 2
Paired-sample t-tests on the differences between learner engagement in RASP and the baselines.

	M	*SD*	95% *CI*	*t*	*df*	Two-Sided *p*
Behavioural engagement	0.29	1.81	(-0.31, 0.89)	0.969	36	0.339
Cognitive engagement	0.33	1.64	(-0.21, 0.88)	1.239	36	0.223
Emotional engagement	−0.12	1.94	(-0.76, 0.53)	−0.367	36	0.716
Social engagement	0.34	1.87	(-0.28, 0.97)	1.111	36	0.274

indicated very strong statistical power (>0.88), and independence of predictors was confirmed (Durbin-Watson values between 1 and 3). Specifically (Table 4), more positive task perceptions significantly predicted higher behavioural ($B = 0.57$, $t = 4.98$, $p < .001$), cognitive ($B = 0.40$, $t = 3.38$, $p = .002$), emotional ($B = 0.83$, $t = 7.60$, $p < .001$), and social engagement ($B = 0.39$, $t = 2.27$, $p = .029$) in RASP (Fig. 10).

Triangulating these quantitative findings, interviewees confirmed that positive learner affection significantly enhanced their engagement in RASP. One interviewee commented, “I felt relaxed and confident when practising English with NAO, so I could engage in the practice for longer … The dialogues with NAO closely resembled authentic IELTS speaking tests, which is very useful. So I worked hard in the [RASP].”

Notably, some interviewees reported that their improved affection in RASP positively influenced their engagement in speaking practice outside the classroom. One participant stated, “I felt overall more comfortable talking with NAO than my classmates … so I practiced speaking harder (in RASP compared to conventional speaking practices).” One learner noted feeling “more comfortable and motivated” to engage in peer-based EFL practice after participating in RASP.

Conflicting with the PLS-SEM results, interviewees also perceived that their enhanced affection and engagement in RASP contributed to speaking proficiency improvement. One participant commented, “RASP was helpful. I concentrated deeply … I learned many new words and became better at generating and expressing ideas smoothly.” However, another interviewee mentioned being more engaged with NAO itself than with the academic content, suggesting that high engagement might not have directly contributed to improvements in speaking proficiency, stating that: “I focused on talking with NAO, not the practice.”

5. Discussions

5.1. Learner affection in RASP

Our statistical data and interview results indicate that learners generally held positive affection in RASP, consistent with previous robot-assisted language learning research (Lin et al., 2022; van den Berghe et al., 2019). Notably, learners reported significantly greater social comfort in RASP compared to conventional speaking practices, primarily due to RASP's more supportive and relaxed environment. Our interviewees appreciated NAO's non-judgemental attitudes, perceived relatedness with NAO, and described it as “like a friend”, which largely improved their affective experience during speaking practice. Our findings extend Zhang et al. (2026) arguments for the positive impacts of non-academic socialising with learning agents on language learner affection. It also support Lee and Lee's (2022) argument for the usefulness of robots in enacting mental immediacy, creating supportive learning environments, and improving learner emotions. Based on these findings, we recommend that researchers and educators integrate robots into EFL speaking practices to foster supportive environments and enhance learner affection.

Moreover, our participants contributed their relaxed experiences and positive affection in RASP to NAO's immediate and positive feedback. Interviewees identified fear of negative evaluation as a major source of social anxiety during speaking activities. However, in RASP, learners felt liberated from such fears, as NAO exclusively offered encouragement and praise, leading to increased relaxation, confidence, and motivation. Also observed by Alemi et al. (2017), this phenomenon aligns with self-determination theory (Deci & Ryan, 2012), which posits that praise enhances learners' awareness of their progress and competence, thus reducing anxiety and improving affection. Our findings show that students value robots' feedback in speaking practices and can develop awareness of progress and experience reduced anxiety due to immediate and positive robot feedback. These findings indicate learners' perceptions of robots' social role in speaking practice, consistent with prior observations (van den Berghe et al., 2019).

Another contributing factor to learners' positive affection in RASP, particularly regarding social comfort, was decreased self-consciousness and reduced awareness of peer judgement. Prior studies have established a negative correlation between self-consciousness and social comfort (Bortolan, 2022; Higa et al., 2008). Supporting these findings, our interviewees indicated that negative affection in conventional speaking practices stemmed from heightened self-consciousness and concern about peer evaluation. In contrast, learners engaged in RASP reported diminished self-awareness, as their focus shifted entirely to skill improvement. This shift significantly enhanced their social comfort and overall affection. This phenomenon aligns with flow theory (Csikszentmihalyi et al., 2014), which argues that real-time interactions and immediate feedback can induce states of intense concentration, intrinsic enjoyment, reduced self-consciousness, and diminished awareness of distractions, such as peer judgement, in learning. It is possible that robots trigger learners’ flow states in RASP, decrease their self-consciousness and awareness of peer judgement, and thereby improve their affection. So far, analysis of robot-assisted language learning from the perspective of the flow theory remains limited, which appears a promising direction for future research.

Despite these positive aspects, we found no statistically significant differences between RASP and conventional speaking practices concerning task perceptions and EFL speaking self-efficacy. The significant relationship identified by our PLS-SEM analysis between

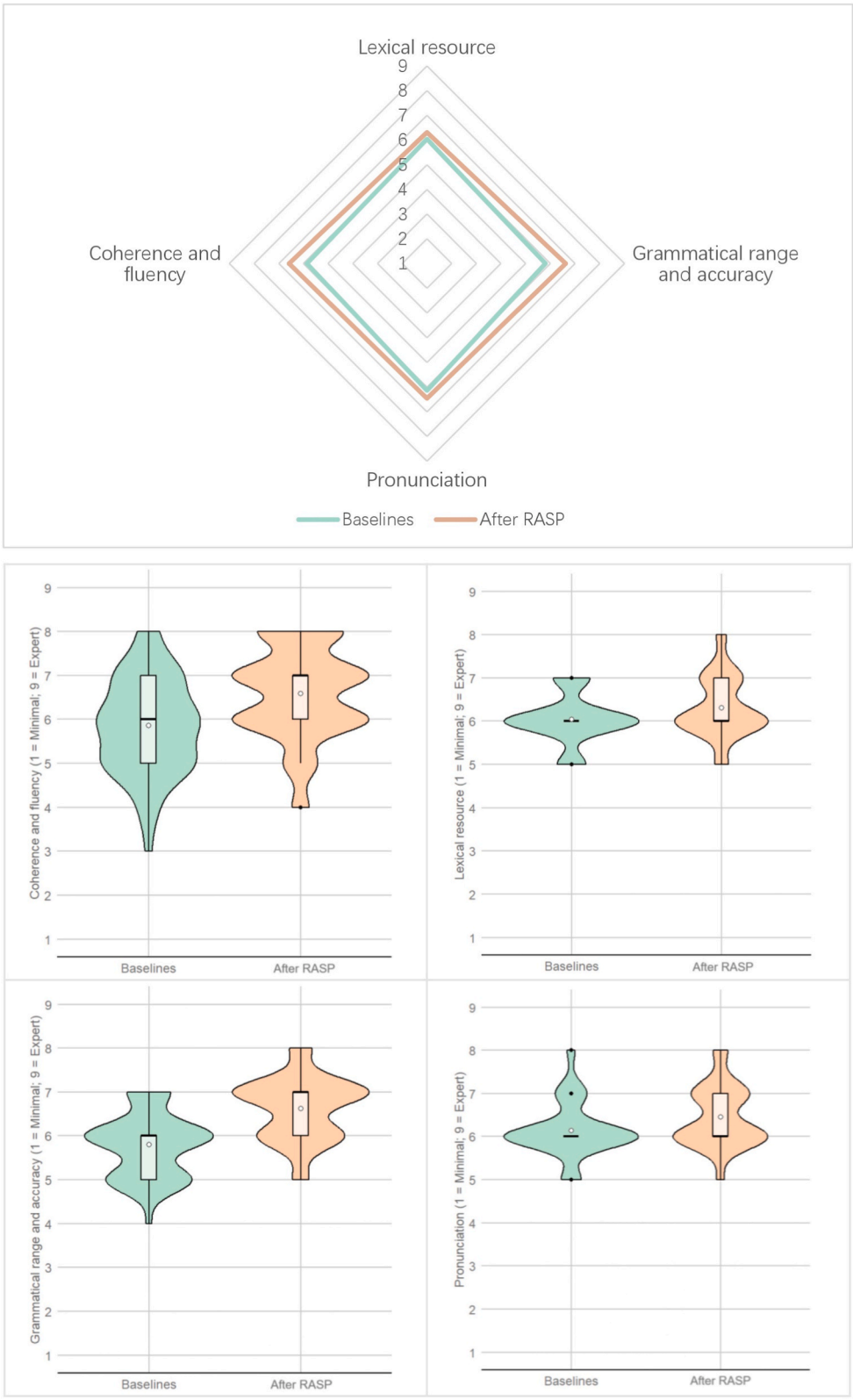

Fig. 8. RASP outcomes in terms of EFL speaking proficiency compared against their baselines.

Table 3
Paired-sample t-tests on the differences between RASP outcomes and learner baselines in terms of EFL speaking proficiency.

	M	*SD*	95% *CI*	*t*	*df*	Two-Sided *p*
Coherence and fluency	−0.75	0.81	(0.51, 0.99)	6.398	47	<0.001**
Lexical resource	−0.29	0.74	(0.08, 0.51)	2.721	47	0.009*
Grammatical range and accuracy	−0.85	0.65	(0.67, 1.04)	9.077	47	<0.001**
Pronunciation	−0.33	0.63	(0.15, 0.52)	3.665	47	<0.001**

*$p < .05$; **$p < .001$.

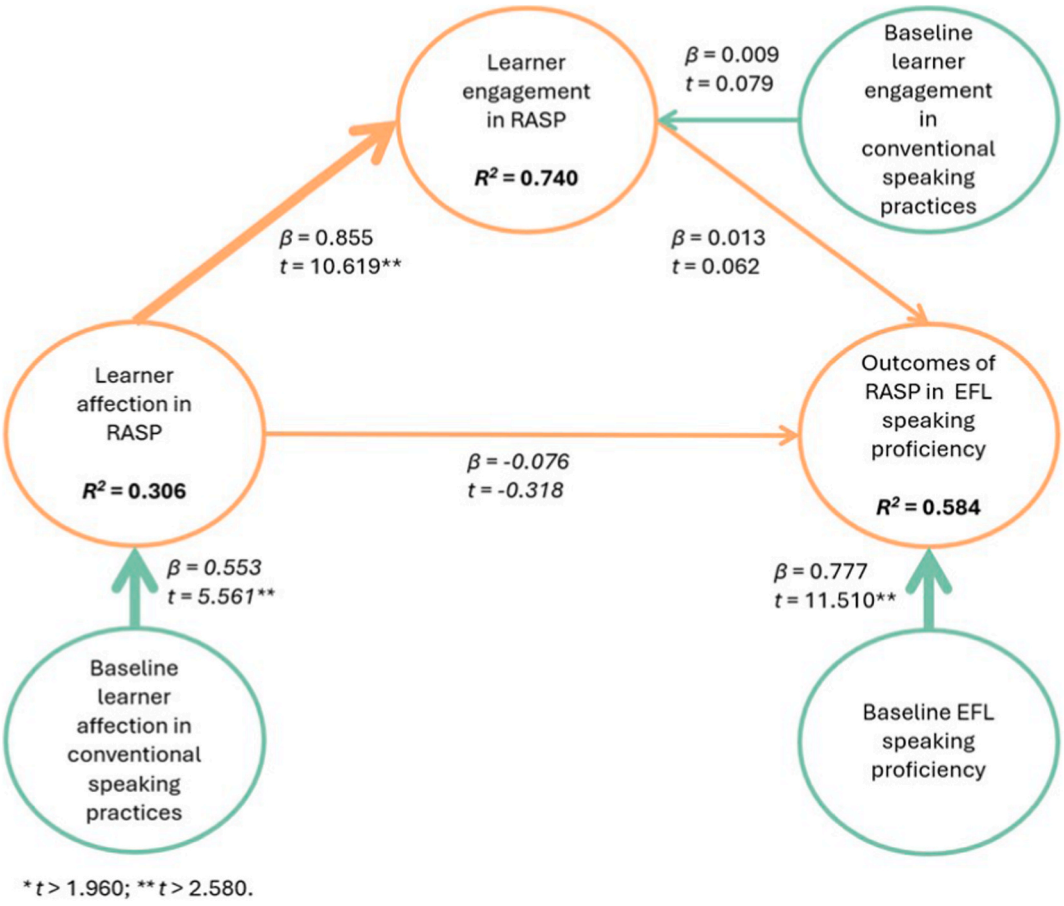

Fig. 9. Summary of the working model.

baseline affection in conventional speaking practices and affection in RASP suggests the stability of language learners' affection across different contexts, echoing Zhang et al.'s (2024) observations regarding null results in technology-enhanced language learning research. Furthermore, technical limitations of NAO, such as inadequate speech recognition and low-quality synthesised voice, negatively affected learners' experiences and affection in RASP, corroborating Lin et al.'s (2022) findings. Thus, future RASP implementations should consider improved speech recognition technology and voice quality to enhance learner affection.

5.2. Learner engagement in RASP

Our data and interview findings validate the high learner engagement with RASP, particularly in terms of behavioural engagement, aligning with earlier robot-assisted language learning research (Lee & Lee, 2022; Randall, 2019). A significant factor driving this engagement was learners' initial curiosity and interest in novel learning technologies, supporting intrinsic motivation theory (Deci & Ryan, 2013), which posits that engagement increases when driven by internal factors like curiosity. However, we also observed a sharp decline in curiosity and interest after two to three sessions as familiarity with the technology increased, providing empirical support for previously reported novelty effects in robot-assisted language learning (van den Berghe et al., 2019).

Another contributor to high engagement was learners' perception of being heard and valued in RASP. Interviewees reported that

Table 4
Parameter estimates of the relationships between learner affection and engagement in RASP.

Models		*B*	SE (*B*)	*β*	*t*	Sig.
Predictors	Outcomes					
(Constant)	Behavioural engagement in RASP	4.124	0.943		4.371	<0.001**
Social comfort in RASP		−0.130	0.078	−0.194	−1.666	0.104
EFL speaking self-efficacy in RASP		−0.007	0.132	−0.007	−0.055	0.957
Task perceptions in RASP		0.569	0.114	0.694	4.981	<0.001**
(Constant)	Cognitive engagement in RASP	3.466	0.986		3.514	0.001*
Social comfort in RASP		−0.094	0.082	−0.148	−1.149	0.257
EFL speaking self-efficacy in RASP		0.174	0.138	0.191	1.267	0.213
Task perceptions in RASP		0.404	0.119	0.520	3.384	0.002*
(Constant)	Emotional engagement in RASP	1.589	0.900		1.765	0.085
Social comfort in RASP		−0.124	0.075	−0.146	−1.667	0.104
EFL speaking self-efficacy in RASP		0.015	0.126	0.013	0.122	0.904
Task perceptions in RASP		0.829	0.109	0.798	7.600	<0.001**
(Constant)	Social engagement in RASP	2.418	1.416		1.708	0.096
Social comfort in RASP		−0.111	0.117	−0.140	−0.947	0.350
EFL speaking self-efficacy in RASP		0.259	0.198	0.227	1.312	0.197
Task perceptions in RASP		0.390	0.171	0.401	2.273	0.029*

*$p < .05$; **$p < .001$.

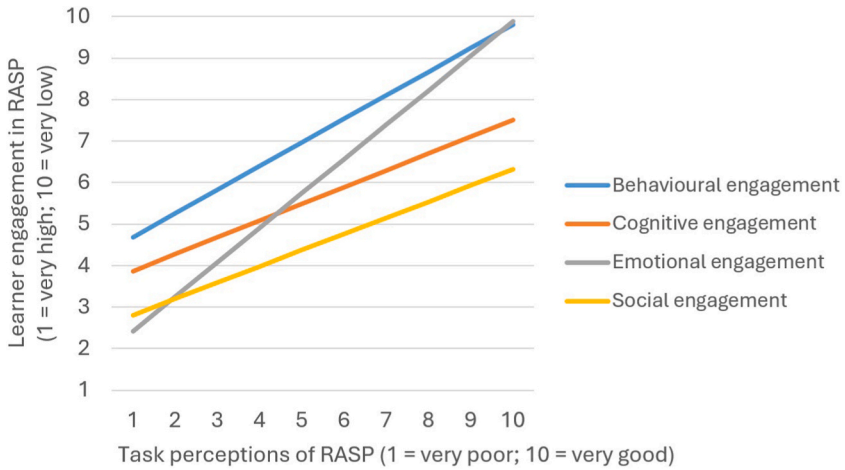

Fig. 10. Impacts of task perceptions of RASP on learner engagement in RASP.

NAO actively listened to their responses and provided immediate, multimodal feedback through gestures, voice, and body language. Such responsive interactions fostered a sense of being understood and valued, an experience rarely found in conventional self-regulated or peer-based speaking practices, enhancing learner engagement, as admitted by our interviewees. These findings resonate with the person-centred approach (Greene, 2017), which emphasises the importance of learners feeling valued as a key driver of motivation and engagement. This phenomenon can also be explained through Self-Determination Theory (Deci & Ryan, 2012), which emphasises that fulfilling learners' basic psychological needs, autonomy, competence, and relatedness, enhances intrinsic motivation and engagement. Feeling valued and heard satisfies the need for relatedness and supports autonomy, fostering ownership and accountability. Consequently, students become more committed to their learning goals, as they perceive their contributions as meaningful and their learning as personally relevant. Thus, our study demonstrates that RASP can effectively engage learners by fostering a sense of recognition and accountability.

However, our analysis revealed no statistically significant differences between RASP and conventional speaking practices in terms of learner engagement, contradicting some robot-assisted language learning studies (e.g., Shin & Shin, 2015). One possible explanation is that in our study, although NAO emulated face-to-face speaking practice using rich body movements and gestures, it did not trigger learners to use body movements and gestures as well. Instead, most of our learners remained largely still during RASP. Hence, unable to trigger their bodily actions sufficiently, NAO may not enable deep cognitive engagement, according to the embodied learning theory (Khazaie & Derakhshan, 2024; Pruss, 2022). So far, research in this direction remains limited. We call for further research into strategies for eliciting greater learner bodily engagement in RASP.

Another possible explanation for the insignificant differences in learner engagement lies in the distinct mechanisms underlying engagement in RASP versus conventional speaking practices. Our PLS-SEM results indicated no significant influence of learner engagement in conventional speaking practices on engagement in RASP, suggesting that learners engage through different

mechanisms in RASP compared to traditional activities. This phenomenon was noted by Zhang et al. (2024) who argued against analysing learner engagement and behaviours in different learning settings in the same way. Given that no prior research has explicitly investigated these distinct engagement patterns, this area merits further exploration.

5.3. RASP outcomes in terms of EFL speaking proficiency

Our quantitative results and interview data demonstrate significant improvements in learners' EFL speaking outcomes after RASP, even when controlling for baseline proficiency. These findings align with previous research (Lin et al., 2022), providing empirical support for the effectiveness of RASP. Interviewees attributed this effectiveness to NAO's role as a professional interlocutor, a role often absent in conventional speaking practices. NAO was perceived as a professional interlocutor "like a real English tutor." Interacting with NAO, learners perceived a strong sense of authenticity and formality of the speaking practice, which motivated increased effort and persistence. These findings echo Lee et al. (2011). Our results also support previous insights regarding learners anthropomorphising robots and assigning them social roles in educational contexts (van den Berghe et al., 2019). Furthermore, our study aligns with the sociocultural theory (Vygotsky, 1978), which highlights the importance of communication with an attentive, professional interlocutor for enhancing language learning outcomes.

Additionally, NAO's multimodal approach to knowledge delivery contributed to improved speaking proficiency. Interviewees found NAO's vivid presentation of language knowledge, through gestures, body movements, and voice, effective for language acquisition. This finding corroborates previous studies highlighting educational robots' ability to give linguistic scaffolding effectively via multisensory communications (Lin et al., 2022; Randall, 2019). Hence, our study reinforces the pedagogical advantages of robots in language education (Lee & Lee, 2022).

5.4. Interplay among learner affection, engagement, and outcome of RASP

Our PLS-SEM analysis reveals significant positive effects of learners' affection, particularly task perceptions, on their engagement in RASP. Interview data corroborate these findings, reinforcing expectancy-value theory (Wigfield & Eccles, 2000) and the theory of planned behaviour (Ajzen, 2005). They also extend the affective filter theory and prior research on the positive influence of learner affection in EFL speaking education (Nagle, 2021) to robot-assisted contexts. Future research aimed at enhancing learner engagement in RASP may focus on improving affection in this learning approach.

However, not all findings are in line with previous EFL speaking studies (e.g., Zeinivand et al., 2015). Our PLS-SEM analysis showed no significant impact of learner affection and engagement in RASP on EFL speaking outcomes. This may be because learners, enjoying high social comfort, became engrossed in entertaining social interactions with the robot and peers, paying insufficient attention to linguistic development. Our interview data support this, with some participants admitting they focused more on interacting with NAO rather than on language learning in RASP. Also observed in previous technology-enhanced language learning studies (Ke et al., 2016; Philp et al., 2010; Zhang et al., 2024), such misalignment of learner attention and interest with their academic goals can lead to limited language learning progress, despite learners' positive affection and deep engagement in the practice. Hence, future RASP practitioners should carefully monitor whether engaged learners focus on language development or become distracted by social interactions, to ensure effective implementation.

Another possible explanation is the discrepancy between learners' self-reported and actual engagement. Heemskerk and Malmberg (2020) found that students often underestimated their engagement in self-report surveys, particularly their cognitive and behavioural engagement. Similarly, our participants might have inaccurately assessed their own engagement on questionnaires, potentially leading to the statistically insignificant relationships. Future research should adopt mixed-methods approaches for a more comprehensive evaluation of learner engagement in RASP.

Moreover, learners' improved affection in RASP may have indirectly improved speaking proficiency by increasing engagement in speaking practice beyond RASP. Our quantitative and qualitative data revealed that after experiencing RASP, our learners had better affection and deeper engagement in self-regulated and peer-based speaking practice outside class. Such extracurricular engagement may have contributed more to EFL speaking development than the eight-session RASP intervention, potentially explaining the insignificant direct effect of learner affection and engagement in RASP on speaking outcomes. Therefore, future studies should consider learner affection and engagement in out-of-class speaking activities to better understand their impact on RASP outcomes.

6. Implications

Our study contributes to the theories of robot-assisted language learning in several respects. First, our RASP targets speaking proficiency development among intermediate-level young adults in higher education and demonstrates overall effectiveness. Although earlier research reported mixed results regarding robots' effectiveness in developing speaking skills (Lee & Lee, 2022) and suggested younger learners benefit more from robot-assisted language learning than adults (Derakhshan et al., 2024), our study provides quantitative and qualitative evidence for robots' significant effectiveness in improving speaking proficiency among adult learners, even after controlling for the novelty effect and baseline factors. Our findings also indicate that RASP may enhance learner affection and engagement in extracurricular speaking practices, both self-regulated and peer-based, potentially supporting long-term EFL speaking development. Hence, educators are recommended to implement RASP to improve EFL speaking skills in the long term among adult learners at the higher educational level.

Second, we found a limited impact of RASP on learner affection and engagement, contrary to prior claims (Derakhshan et al., 2024;

Randall, 2019). By examining RASP dynamics through both objective and subjective measures, we found that learner affection and engagement across multiple dimensions initially increased, then declined before stabilising. Notably, emotional engagement exhibited a slight decrease over the course of RASP, likely due to the diminishing novelty effect after two to three sessions. We also identified the hard-to-change nature of learner characteristics and learners' insufficient bodily actions in RASP, which further constrained the positive effects of this learning method on affection and engagement. Our findings underscore the complex, multidimensional, and dynamic nature of learner affection and engagement within robot-assisted language learning contexts, which future research should carefully consider. Based on these insights, we also recommend that educators pay particular attention to fostering and sustaining learner affection and engagement when implementing RASP.

Third, our study underscores the usefulness of activity theory in analysing RASP. Through this theoretical lens, we identified the dynamic between learners (*Subject*), the speaking practice task (*Object*), and robot and peer interlocutors (*Community*). We illuminate variations in learner affection across different elements of RASP. We found that robot assistance can substantially enhance learner affection towards interlocutors in speaking activities (*Community*), while learners' affection towards themselves (*Subject*) and the practice task (*Object*) remained relatively stable. Building on these findings, educators may consider employing RASP to engage learners with high social anxiety in speaking practice. Our results thus support Lin et al.'s (2022) argument for the high adaptability of robot-assisted learning across diverse language learning theories. With reference to our conceptual framework, researchers are encouraged to leverage activity theory further to explore aspects such as learner engagement with various elements of RASP. It is important to note that our study focuses primarily on the three central elements of activity theory, *Subject*, *Object*, and *Community*, while the expanded versions of the theory comprise a more complex, dynamic network that includes additional elements, such as artefacts, rules, and division of labour (Engeström, 2001b). Future research could further refine our conceptual framework of RASP and extend the analysis by incorporating these additional elements, thereby achieving a more comprehensive understanding of this learning approach.

Our study has also provided rich practical implications by illuminating robots' strong potential as qualified interlocutors in speaking practice, a role that is crucial yet often insufficiently fulfilled in real-life EFL settings (Bailey, 2003). With voice synthesis and recognition systems, robot interlocutors can assign academic tasks, ask questions, provide feedback, and deliver linguistic scaffolding, aligning with prior expectations regarding educational robots' potentials (Deng et al., 2024; Lin et al., 2022). Compared to other technologies like multimedia, animated characters, and chatbots, robots, particularly anthropomorphic ones like NAO, afford the physical presence and non-verbal, multisensory communicative cues, thereby having more capability of emulating human face-to-face interactions and adopting natural social roles (van den Berghe et al., 2019). These features may lead learners to experience a stronger sense of "real" with robot interlocutors, as reflected in interviewees' descriptions of NAO as "like a friend" and "close to authentic IELTS speaking examiners." Consequently, robot interlocutors can create a strong sense of authenticity in speaking practice (van den Berghe et al., 2019), which is essential for the practice effectiveness in developing speaking proficiency (Lin et al., 2022). Therefore, robots may be more useful than other technologies in serving as the interlocutor in speaking practice, contributing to better EFL speaking development.

Robot interlocutors also hold some advantages over human interlocutors in speaking practice. At the linguistic level, robot interlocutors are perceived as "very professional" for offering high-quality language input via feedback and scaffolding, according to our interview results. Such high-quality language input is the prerequisite for effective speaking practice, aligning with Schmidt's (2012) and Ellis's (2019) argument. Hence, compared to peer interlocutors who generally lack language proficiency (Dao, 2020), robot interlocutors appear better in facilitating EFL speaking development. At the emotional level, our finding showed robot interlocutors' usefulness in bridging mental and physical immediacy with learners, creating a comfortable, supportive environment for speaking practice, and thus improving learner affection and engagement, consistent with previous arguments (Lee & Lee, 2022; Randall, 2019). In contrast, peer interlocutors and teacher interlocutors can cause strong self-awareness and anxiety about judgement and negative feedback in learners, thus reducing learner affection and engagement in speaking practice (Archbell & Coplan, 2022). Therefore, robot interlocutors may foster greater learner affection and engagement compared to peer or teacher interlocutors, a distinction that is especially significant given the crucial role of positive affect (Mahripah, 2013) and deep engagement (Ding & Zhu, 2025) in effective EFL speaking practice.

Nonetheless, robot interlocutors remain far from perfect. Our interviews revealed NAO's inadequate speech recognition and low-quality synthesised voice, which negatively impacted learners' experiences and affection towards RASP. NAO's limited speech recognition also hampers its ability to identify and correct learners' speech errors, a critical function in speaking education (Gumbaridze, 2013). NAO' low-quality synthesised voice may reinforce mispronunciations and hinder language comprehension (In & Han, 2015). Such technical limitations likely constrained RASP's effectiveness in developing EFL speaking proficiency. As inadequate speech recognition and low-quality synthesised voice are common issues among educational robots (Lin et al., 2022), our findings regarding NAO's limitations may generalise across RASP contexts and robot types. Future practitioners should take these constraints into consideration. To address the limitations, we suggest employing an advanced speech recognition system and AI-generated robot voices to enhance learners' experiences and efficiency in RASP.

Moreover, our interviewees reported difficulties processing language knowledge alongside real-time interactions with the robot interlocutor in RASP, a phenomenon frequently observed in EFL speaking practice (Gomez, 2021). Providing knowledge notes outlining key language concepts during RASP may help address the problem, which can scaffold learners during speaking practice and serve as takeaways for further study (Mirahmadi & Alavi, 2016). Hence, we recommend that educators provide EFL learners with knowledge notes during RASP, allowing learners to review and process new language knowledge at their own pace and enhancing both the efficiency and effectiveness of RASP.

Overall, our study demonstrates the significant effectiveness of robots as interlocutors in supporting EFL speaking practice,

surpassing other technologies and human interlocutors, and highlights their considerable potential in language education more broadly. In light of the urgent global challenge of teacher shortages (Symeonidis et al., 2025), we recommend that policymakers endorse the integration of robotics and related educational tools and materials into language curricula as a viable solution. To maximise the efficiency of RASP and other robot-assisted learning implementations, we further advise the inclusion of professional development programmes that prepare language teachers to effectively design, implement, and manage robot-assisted learning environments. Additionally, we suggest prioritising funding for advanced educational robots equipped with features such as high-quality voice synthesis and speech recognition, which may further enhance the effectiveness of RASP and similar applications, in line with Lin et al.'s (2022) recommendations.

7. Conclusions and limitations

This study proposed a RASP framework grounded in activity theory, the affective filter theory, embodied learning theory, expectancy-value theory, the noticing hypothesis, and engagement theories. We investigated the process and effects of this learning method using a mixed-methods approach. Findings revealed positive learner affection, deep engagement, and satisfactory EFL speaking proficiency outcomes of RASP. While affection significantly enhanced engagement, neither affection nor engagement had a statistically significant impact on learning outcomes. These findings were interpreted through self-determination theory, flow theory, sociocultural theory, person-centred learning theory, and expectancy-value theory.

Despite triangulation of subjective and objective data, our findings may not be generalised across samples. This study examined a single cohort of Chinese EFL students from one higher educational institution who had overall intermediate language proficiency and experience with conventional EFL speaking practice and technology-enhanced learning but no prior exposure to robot-assisted language. This highly homogeneous cohort, although reflective of learners sharing comparable backgrounds, restricted the examination of factors such as target language, digital literacy, prior training, educational history, and cultural identity. These factors may influence the experiences and outcomes of RASP, as individuals' situational, developmental, and cultural contexts play a considerable role in language learning processes (Horwitz, 1999). For instance, our participants demonstrated upper-intermediate levels of engagement in RASP, not significantly differing from that in conventional speaking practice. This finding may not be extended to elementary and secondary EFL education since younger students generally exhibit greater willingness to engage with robot-assisted learning (Randall, 2019). Additionally, students with different sociocultural backgrounds may adhere to different social norms, resulting in varied levels and forms of engagement in RASP. Researchers observed that compared to those with western cultural backgrounds, Chinese EFL students tend to speak less and initiate conversations less frequently in interactions (Li & Jia, 2006; Zheng, 2021). By this logic, learners with western cultural backgrounds may have deeper engagement in RASP than our participants. To deepen the understanding of RASP and broaden the scope of research in this area, future studies should consider a larger, more diverse population with various situational, developmental, and cultural backgrounds when investigating learner affection, engagement, and outcomes of RASP.

Researchers and educators may also be particularly mindful of robot types when applying our findings. Drawing on the literature (e.g., Randall, 2019; Shin & Shin, 2015), we selected the NAO robot for our study due to its capabilities in physical interactions and non-verbal social cues, which closely align with the requirements of our RASP framework. However, given the considerable variation in design and functionality among different robot models (Randall, 2019), our results may not be directly generalisable to RASP implementations involving other robots. For instance, interviews revealed that some students reported increased engagement through eye contact with NAO, a feature absent in certain robots such as Scribble, which lack LED eye-gaze capabilities. Consequently, RASP implementations utilising robots without such eye-gaze functionality may not elicit the same level of learner engagement observed with NAO. Moreover, NAO's programming environment, Choregraphe, does not support adaptive feedback or personalised learning pathways, whereas other platforms, such as the pocket robot partner described by Yorita et al. (2013), offer these advanced functionalities. Considering the documented benefits of adaptive feedback and personalised learning in enhancing the efficacy of robot-assisted language learning (Randall, 2019), RASP implementations using robots with these capabilities may yield superior EFL speaking outcomes compared to NAO-based interventions. Similarly, recent advancements have seen an increasing number of educational robots equipped to recognise learner behaviours and emotions and respond with appropriate feedback (Schiavo et al., 2024; Zhao et al., 2025), features beyond the current scope of NAO. Given the significant impact of behaviour and emotion recognition on improving learner affection in EFL speaking practice (Chen & Lee, 2011), this limitation may have constrained the effectiveness of RASP in our study. Future research should explore RASP employing diverse robot types and systematically compare their effectiveness to deepen understanding of this learning approach and improve its generalisability across varied technological contexts.

Our study has limitations. First, it focused solely on EFL speaking proficiency as the outcome of RASP, while other skills and knowledge may also be developed. Interviewees reported improvements in idea generation and organisation from RASP, which warrant further exploration in future studies. Second, we relied heavily on self-reported data for learner affection and engagement. While valuable, objective measures such as structured observations (Heemskerk & Malmberg, 2020) or eye-tracking data (Zhang et al., 2025a) could provide a more comprehensive understanding. Future research should integrate both self-reported and objective data in exploring learner affection and engagement in RASP.

Robot-assisted learning is gaining popularity across disciplines. This study explored the use of robots in developing EFL speaking among undergraduates, offering insights into its potential in language education and beyond. To further unlock the capabilities of this learning method, we encourage research integrating various state-of-the-art technologies into robot-assisted learning of EFL speaking. For example, adaptive feedback and personalised learning pathways, features not supported by NAO's official software in our study, may enhance the effectiveness of robot-assisted learning (Randall, 2019). Future research could further investigate the potential of RASP by incorporating these functionalities into robot-assisted language learning. Additionally, Khazaie and Derakhshan (2024)

integrated augmented reality technology into robot-assisted learning of EFL listening and reading, enabling learners to visualise an educational robot in various medical scenarios via mobile devices. Their results showed that the integration of augmented reality improved the effectiveness of robot-assisted language learning by enhancing students' comprehension. This impact may be extended to EFL speaking development. Artificial intelligence technology also appears promising in this direction. For example, Derakhshan et al. (2025) incorporated Artificial intelligence technology into robot-assisted learning of EFL speaking for medical purposes, where Artificial Intelligence analysed and evaluated students' speaking quality and directed robot responses. Considering the great potential of Artificial intelligence technology, future research may further leverage it, for instance, by generating personalised conversational content for real-time, free-flow learner-robot interactions.

CRediT authorship contribution statement

Ruofei Zhang: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Di Zou:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Haoran Xie:** Validation, Supervision, Software, Resources, Project administration, Methodology, Conceptualization. **Gary Cheng:** Visualization, Validation, Supervision, Software, Resources, Project administration.

Ethics declarations

I confirm that all the research meets ethical guidelines and adheres to the legal requirements of the study country.

Declaration of generative AI generative-ai

Generative AI and AI-assisted technologies (ChatGPT) have only been used to check the grammar of the manuscript.

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Conflict of interest

None.

Appendix A. Learner affection in RASP

To what extent do the following statements reflect your actual situation? Please rate them on a scale from 1 to 10: 1 = Completely untrue, not related to me at all 10 = Completely true, exactly like me	
Social comfort (adapted from Connor et al., 2001)	
1. In RASP, fear of embarrassment causes me to avoid interacting with classmates.*	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
2. In RASP, I avoid to be the center of attention.*	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
3. In RASP, being embarrassed or looking stupid are among my worst fears.*	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
EFL speaking self-efficacy (adapted from Gan et al., 2022)	
4. I can speak English fluently.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
5. I can communicate with NAO and classmates in English.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
6. I can try to keep a high level of self-confidence in RASP.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
7. I can understand and participate in the most complex, difficult questions in RASP.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
8. I can understand the most complex, difficult speech by my classmates and NAO in speaking practices.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
9. I can master the speaking skills and language knowledge learned and practiced in RASP.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
10. I can NOT use the speaking skills and language knowledge learned and practiced in RASP for real-life or authentic IELTS tests.*	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
11. I can do very well in RASP.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]

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(*continued*)

To what extent do the following statements reflect your actual situation? Please rate them on a scale from 1 to 10: 1 = Completely untrue, not related to me at all 10 = Completely true, exactly like me	
12. I can do a good job of participating in RASP fully in English.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
Task perceptions (adapted from Ghani et al., 2019*)*	
13. Developing English proficiency through RASP is a good idea.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
14. I feel positive towards RASP.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
15. I believe that RASP help me to be more engaged in developing English proficiency.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
16. I generally favour RASP for developing English proficiency.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
17. I believe that it is a good idea for me to have RASP for my future English study.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]

* Reversed item.

Appendix B. Learner engagement in RASP

To what extent do the following statements reflect your actual situation? Please rate them on a scale from 1 to 10: 1 = Completely untrue, not related to me at all 10 = Completely true, exactly like me	
Behavioural engagement (adapted from Skinner et al., 2009)	
1. In RASP, I engage in each question very carefully.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
2. I try hard to speak well in RASP.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
3. In RASP, I DON'T work as hard as I can.*	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
Cognitive engagement (adapted from Skinner et al., 2009*)*	
4. I try to make different knowledge points and speaking strategies fit together and use them in RASP.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
5. I try to relate new language skills and knowledge to what I already know and use them in RASP.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
6. In RASP, I make up my own example answers to help me master important speaking skills and knowledge points.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
Emotional engagement (adapted from Deng et al., 2020*)*	
7. I was inspired to further develop my English proficiency in RASP.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
8. I found RASP interesting.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
9. I enjoyed learning, speaking, and listening English in RASP.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
Social engagement (adapted from Deng et al., 2020*)*	
10. I often responded to my classmates and NAO in EFL speaking practices.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
11. In RASP, I regularly discuss with my classmates on knowledge points and speaking skills.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]
12. In RASP, I shared learning materials (e.g., notes, vocabulary definitions, grammar knowledge) with my classmates.	[1] [2] [3] [4] [5] [6] [7] [8] [9] [10]

*Reversed item.

Appendix C. Indicators of the validity and reliability of the PLS-SEM analysis

Table 1
Measurement of RASP model

Level one	Level two	Measures
Learner affection in RASP	Social comfort in RASP	Post-questionnaire results on "Social comfort"
	EFL speaking self-efficacy in RASP	Post-questionnaire results on "EFL speaking self-efficacy"
	Task perceptions in RASP	Post-questionnaire results on "Task perceptions"
Learner engagement in RASP	Behavioural engagement in RASP	Post-questionnaire results on "Behavioural engagement"
	Cognitive engagement in RASP	Post-questionnaire results on "Cognitive engagement"
	Emotional engagement in RASP	Post-questionnaire results on "Emotional engagement"
	Social engagement in RASP	Post-questionnaire results on "Social engagement"

(*continued on next page*)

Table 1 (*continued*)

Level one	Level two	Measures
Learning outcomes of RASP in EFL speaking proficiency	Coherence and fluency after RASP	Post-test results on "Coherence and fluency"
	Lexical resource after RASP	Post-test results on "Lexical resource"
	Grammatical range and accuracy after RASP	Post-test results on "Grammatical range and accuracy"
	Pronunciation after RASP	Post-test results on "Pronunciation"
Baseline learner affection in conventional speaking practices	Baseline social comfort in conventional speaking practices	Pre-questionnaire results on "Social comfort"
	Baseline EFL speaking self-efficacy in conventional speaking practices	Pre-questionnaire results on "EFL speaking self-efficacy"
	Baseline task perceptions in conventional speaking practices	Pre-questionnaire results on "Task perceptions"
Baseline learner engagement in conventional speaking practices	Baseline behavioural engagement in conventional speaking practices	Pre-questionnaire results on "Behavioural engagement"
	Baseline cognitive engagement in conventional speaking practices	Pre-questionnaire results on "Cognitive engagement"
	Baseline emotional engagement in conventional speaking practices	Pre-questionnaire results on "Emotional engagement"
	Baseline social engagement in conventional speaking practices	Pre-questionnaire results on "Social engagement"
Baseline EFL speaking proficiency	Baseline coherence and fluency	Pre-test results on "Coherence and fluency"
	Baseline lexical resource	Pre-test results on "Lexical resource"
	Baseline grammatical range and accuracy	Pre-test results on "Grammatical range and accuracy"
	Baseline pronunciation	Pre-test results on "Pronunciation"

Table 2
Indicators for the reliability and validity of the PLS-SEM measurement model

Variables	Indicators	Factor loadings	*rhoA*	*AVE*
Learning outcomes of RASP in EFL speaking proficiency	Coherence and fluency after RASP	0.820	0.857	0.689
	Lexical resource after RASP	0.842		
	Grammatical range and accuracy after RASP	0.843		
	Pronunciation after RASP	0.816		
Learner affection in RASP	Social comfort in RASP	0.813	0.826	0.739
	EFL speaking self-efficacy in RASP	0.872		
	Task perceptions in RASP	0.892		
Learner engagement in RASP	Behavioural engagement in RASP	0.867	0.915	0.783
	Cognitive engagement in RASP	0.904		
	Emotional engagement in RASP	0.927		
	Social engagement in RASP	0.840		
Baseline EFL speaking proficiency	Baseline coherence and fluency	0.804	0.820	0.601
	Baseline lexical resource	0.688		
	Baseline grammatical range and accuracy	0.877		
	Baseline pronunciation	0.720		
Baseline learner affection in conventional speaking practices	Baseline social comfort in conventional speaking practices	−0.756	0.780	0.534
	Baseline EFL speaking self-efficacy in conventional speaking practices	0.434		
	Baseline task perceptions in conventional speaking practices	0.712		
Baseline learner engagement in conventional speaking practices	Baseline behavioural engagement in conventional speaking practices	0.554	0.802	0.609
	Baseline cognitive engagement in conventional speaking practices	0.463		
	Baseline emotional engagement in conventional speaking practices	0.763		
	Baseline social engagement in conventional speaking practices	0.757		

Table 3
The heterotrait-monotrait ratio of correlations (HTMT) indicating the discriminant validity of the measurement models of PLS-SEM (to be continued)

	Learning outcomes of RASP in EFL speaking proficiency	Baseline learner engagement in conventional speaking practices	Learner engagement in RASP	Baseline EFL speaking proficiency	Baseline learner affection in conventional speaking practices
Learning outcomes of RASP in EFL speaking proficiency	.				
Baseline learner engagement in	0.747	.			

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Table 3 (*continued*)

	Learning outcomes of RASP in EFL speaking proficiency	Baseline learner engagement in conventional speaking practices	Learner engagement in RASP	Baseline EFL speaking proficiency	Baseline learner affection in conventional speaking practices
conventional speaking practices					
Learner engagement in RASP	0.887	0.606	.		
Baseline EFL speaking proficiency	0.376	0.345	0.193	.	
Baseline learner affection in conventional speaking practices	0.795	0.828	0.732	0.322	.
Learning outcomes of RASP in EFL speaking proficiency	0.229	0.353	0.091	0.880	0.359

Table 4
Variance inflation factors (VIF) indicating the collinearity of the PLS-SEM structural model

Predictors	Outcomes		
	Learner engagement in RASP	Learning outcomes of RASP in EFL speaking proficiency	Learner affection in RASP
Learner affection in RASP	1.543	4.201	n/a
Baseline learner engagement in conventional speaking practices	1.543	n/a	n/a
Learner engagement in RASP	n/a	4.033	n/a
Baseline EFL speaking proficiency	n/a	1.099	n/a
Baseline learner affection in conventional speaking practices	n/a	n/a	1.003

Appendix D. Model estimates of multiple regression

Table 1
Estimates of the model from learner affection in learner engagement in RASP

Models		R^2	R^2_{change}	F_{change}	Sig. F_{change}	Durbin-Watson
Predictors	Outcomes					
Social comfort in RASP EFL speaking self-efficacy in RASP Task perceptions in RASP	Behavioural engagement in RASP	0.663	0.663	25.547	<0.001**	1.767
Social comfort in RASP EFL speaking self-efficacy in RASP Task perceptions in RASP	Cognitive engagement in RASP	0.589	0.589	18.593	<0.001**	2.109
Social comfort in RASP EFL speaking self-efficacy in RASP Task perceptions in RASP	Emotional engagement in RASP	0.808	0.808	54.718	<0.001**	2.380
Social comfort in RASP EFL speaking self-efficacy in RASP Task perceptions in RASP	Social engagement in RASP	0.459	0.459	11.018	<0.001**	2.549

**$p < .001$.

Data availability

The participants of this study did not give written consent for their data to be shared publicly, so due to the sensitive nature of the research supporting data is not available.

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